Normal Targeting at Corank One: A Complete One-Column Criterion

Automated mathematical discovery packet
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Status. Likely valid partial result, subject to human review. The general normal targeting problem remains open; this packet completely treats the subcase rank $X = m - 1$ over $\mathbb{C}$.

Abstract

Given $X, Y \in \mathsf{M}_{m\times n}(\mathbb{C})$, the normal targeting problem asks when a normal $A \in \mathsf{M}_m(\mathbb{C})$ satisfies $AX = Y$. We solve the maximal-rank singular case rank $X = m - 1$. A compact singular value decomposition reduces the problem to completing one prescribed block column $\left[\begin{smallmatrix} B \\ c^* \end{smallmatrix}\right]$ by one free column. Normality is equivalent to a positive rank-one defect identity and a single vector compatibility equation involving only one phase and one complex scalar. When B is normal, the latter condition says exactly that the eigenvalues of B visible from c lie on an affine line. This gives an explicit 4×4 obstruction showing that the positive-semidefinite defect condition alone is not sufficient even when only one column is missing.

1 Source question and scope

Bierly, Garcia, and Horn introduce the linear targeting problem and state in Section 10 that the normal case remains open [1]. The relevant source passage (p. 13) is reproduced below.

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Although the normal targeting problem remains open, the following theorem provides a solution in a special case that provides an alternative solution to the reflection and orthogonal projection targeting problems.

Theorem 10.1. *Let $X, Y \in \mathsf{M}_{m\times n}(\mathbb{C})$ with $m \geq n \geq 1$ and $X \neq 0$. Let $\lambda, \mu \in \mathbb{C}$ be distinct. If $m = n$ and either $Y = \lambda X$ or $Y = \mu X$, assume that* rank $X < m$.

(a) There is a normal $A \in \mathsf{M}_m(\mathbb{C})$ such that the spectrum of A is contained in $\{\lambda, \mu\}$ and $AX = Y$ if and only if $(Y - \lambda X)^*(Y - \mu X) = 0$.

(b) If X, Y, λ, and μ are real and $(Y - \lambda X)^{\mathsf{T}}(Y - \mu X) = 0$, then there is a real normal $A \in \mathsf{M}_m(\mathbb{R})$ such that the spectrum of A is contained in $\{\lambda, \mu\}$ and $AX = Y$.

Proof. (a) ($\Rightarrow$) If $AX = Y$, A is normal, and the eigenvalues of A are λ (with multiplicity $p \geq 1$) and μ (with multiplicity $m - p \geq 1$), then there is a unitary $U \in \mathsf{M}_m(\mathbb{C})$ and a diagonal $\Lambda = \lambda I_p \oplus \mu I_{m-p}$ such that $A = U\Lambda U^*$. Then

The question is:

Characterize those $X, Y \in \mathsf{M}_{m\times n}(\mathbb{C})$ for which there is a normal $A \in \mathsf{M}_m(\mathbb{C})$ satisfying $AX = Y$.

The source solves several structured variants, including normal matrices with two prescribed spectral values, but not the unrestricted problem. We treat the first rank below the invertible case: $\operatorname{rank} X = m-1$. Since the source assumes $m \geq n$, this forces $n \in \{m-1, m\}$.

2 Proof intuition

The columns of X determine the action of A on the $(m-1)$-dimensional space $\operatorname{Ran} X$. After choosing an orthonormal basis adapted to this space, every targeting matrix therefore has all but its last column fixed. Write it as

$$\widehat{A} = \begin{bmatrix} B & d \\ c^* & \eta \end{bmatrix}.$$

Comparing the upper-left and upper-right blocks of $\widehat{A}^*\widehat{A}$ and $\widehat{A}\widehat{A}^*$ gives the whole answer. The upper-left comparison says that the computable defect $B^*B - BB^* + cc^*$ must be dd^*, hence positive semidefinite of rank at most one. If it has rank one, it determines d up to a phase. The upper-right comparison is then one explicit compatibility equation for that phase and η. The bottom-right comparison follows automatically by taking traces.

When B is normal, the defect forces $d = \omega c$. In a spectral basis for B, the compatibility equation says that $\lambda - \omega\overline{\lambda}$ is constant on every eigenvalue seen by c. Geometrically this means precisely that those eigenvalues lie on one line in the complex plane.

3 The corank-one criterion

Let $r = m-1$. Take a compact singular value decomposition

$$X = V_1 \Sigma W_1^*,$$

where $V_1 \in \mathsf{M}_{m\times r}(\mathbb{C})$ and $W_1 \in \mathsf{M}_{n\times r}(\mathbb{C})$ have orthonormal columns and $\Sigma \in \mathsf{M}_r(\mathbb{C})$ is positive diagonal. Complete V_1 to a unitary $V = [\, V_1 \;\, v \,]$. Assuming $\ker X \subseteq \ker Y$, set

$$Z := YW_1\Sigma^{-1}, \qquad V^*Z = \begin{bmatrix} B \\ c^* \end{bmatrix}, \quad B \in \mathsf{M}_r(\mathbb{C}),\ c \in \mathbb{C}^r,$$

and define the Hermitian defect

$$H := B^*B - BB^* + cc^*.$$

Theorem 1 (Complete criterion when $\operatorname{rank} X = m-1$)**.** *Let $X, Y \in \mathsf{M}_{m\times n}(\mathbb{C})$, $m \geq n$, and $\operatorname{rank} X = m-1$. There is a normal $A \in \mathsf{M}_m(\mathbb{C})$ satisfying $AX = Y$ if and only if:*

(i) $\ker X \subseteq \ker Y$*;*

(ii) *H is positive semidefinite and* $\operatorname{rank} H \leq 1$*; and*

(iii) *if* $\operatorname{rank} H = 1$ *and $h \in \mathbb{C}^r$ is any vector with $H = hh^*$, there are $\omega \in \mathbb{T}$ and $\eta \in \mathbb{C}$ such that*

$$B^*(\omega h) + c\eta = Bc + (\omega h)\overline{\eta}. \tag{1}$$

If $H = 0$, condition (iii) *is vacuous. In the rank-one case, every normal targeting matrix is obtained from a solution* (ω, η) *of* (1) *by*

$$A = V \begin{bmatrix} B & \omega h \\ c^* & \eta \end{bmatrix} V^*. \tag{2}$$

When $H = 0$, every $\eta \in \mathbb{C}$ gives a solution by using the same formula with the upper-right vector equal to zero.

Proof. The inclusion $\ker X \subseteq \ker Y$ is necessary for any targeting matrix. It is equivalent to $\operatorname{Ran} Y^* \subseteq \operatorname{Ran} X^* = \operatorname{Ran} W_1$, and hence

$$Y = YW_1W_1^*. \tag{3}$$

Moreover, $AX = Y$ is equivalent to $AV_1 = Z$. Indeed, $AX = Y$ gives this after right multiplication by $W_1\Sigma^{-1}$, while the converse follows from (3):

$$AX = AV_1\Sigma W_1^* = Z\Sigma W_1^* = YW_1W_1^* = Y.$$

Consequently every targeting matrix, and only a targeting matrix, has the form

$$A = V\widehat{A}V^*, \qquad \widehat{A} = \begin{bmatrix} B & d \\ c^* & \eta \end{bmatrix}$$

for some $d \in \mathbb{C}^r$ and $\eta \in \mathbb{C}$.

Direct multiplication gives

$$\widehat{A}^*\widehat{A} = \begin{bmatrix} B^*B + cc^* & B^*d + c\eta \\ d^*B + \overline{\eta}c^* & d^*d + |\eta|^2 \end{bmatrix}$$

and

$$\widehat{A}\widehat{A}^* = \begin{bmatrix} BB^* + dd^* & Bc + d\overline{\eta} \\ c^*B^* + \eta d^* & c^*c + |\eta|^2 \end{bmatrix}.$$

Thus the upper-left blocks agree exactly when

$$H = dd^*, \tag{4}$$

and the upper-right blocks agree exactly when

$$B^*d + c\eta = Bc + d\overline{\eta}. \tag{5}$$

Taking traces in (4), and using $\operatorname{tr}(B^*B - BB^*) = 0$, yields $d^*d = c^*c$; hence the bottom-right blocks then agree automatically. The lower-left equality is the adjoint of (5). Therefore (4) and (5) are necessary and sufficient for normality.

Equation (4) is solvable precisely when $H \geq 0$ and $\operatorname{rank} H \leq 1$. If $H = hh^*$ has rank one, all of its solutions are $d = \omega h$ with $\omega \in \mathbb{T}$, and (5) becomes (1). If $H = 0$, then $d = 0$; taking the trace also gives $c = 0$, and $H = 0$ reduces to $B^*B = BB^*$. Hence (5) is automatic for every η. This proves the criterion and construction. Unitary conjugation by V preserves normality. $\square$

Remark 2 (A three-real-variable test)**.** *Once H has passed the rank-one test, the original m^2 normality equations have been reduced to* (1)*: one unit-modulus phase ω and one complex scalar η. For fixed ω, this is a real-linear system in $\Re\eta$ and $\Im\eta$. Thus the criterion is directly checkable and constructive without optimizing over a full $m \times m$ matrix.*

4 Closed form for a normal compression

Suppose now that B is normal. Let

$$B = U \operatorname{diag}(\lambda_1, \dots, \lambda_r) U^*, \qquad \widehat{c} = U^* c,$$

and call

$$\Lambda_c := \{\lambda_j : \widehat{c}_j \neq 0\}$$

the *active spectrum* of (B, c). Equivalently, an eigenvalue is active when the spectral projection of B at that eigenvalue does not annihilate c; this definition is independent of the chosen basis within repeated eigenspaces.

Corollary 3 (Affine-line criterion)**.** *If B is normal, the prescribed block column $\left[\begin{smallmatrix} B \\ c^* \end{smallmatrix}\right]$ has a normal one-column completion if and only if either $c = 0$ or Λ_c is contained in an affine line in $\mathbb{C}$.*

Proof. For normal B, $H = cc^*$. When $c \neq 0$, Theorem 1 permits $h = c$, so $d = \omega c$ for some $\omega \in \mathbb{T}$. In the spectral basis of B, equation (1) says, for every active index j,

$$\lambda_j - \omega \overline{\lambda_j} = \eta - \omega \overline{\eta}. \tag{6}$$

Write $\omega = e^{2i\theta}$. Multiplication by $e^{-i\theta}$ transforms the left side of (6) into

$$e^{-i\theta} \lambda_j - e^{i\theta} \overline{\lambda_j} = 2i \operatorname{Im}(e^{-i\theta} \lambda_j).$$

It is independent of j precisely when all active eigenvalues have the same orthogonal coordinate relative to direction $e^{i\theta}$, that is, precisely when they lie on an affine line parallel to $e^{i\theta}$. If such a line exists, choose η on the same line; then (6) holds and the construction in Theorem 1 gives a normal completion. The case $c = 0$ is immediate. $\square$

5 A sharp four-dimensional obstruction

Proposition 4. *Let*

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \qquad Y = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & i \\ 1 & 1 & 1 \end{bmatrix}.$$

Then $\operatorname{rank} X = 3$*,* $\ker X \subseteq \ker Y$*, and the source paper's necessary defect matrix $H(B, C)$ is positive semidefinite of rank one, but there is no normal $A \in \mathsf{M}_4(\mathbb{C})$ with $AX = Y$.*

Proof. Here $V = I_4$, $\Sigma = I_3$,

$$B = \operatorname{diag}(0, 1, i), \qquad c = (1, 1, 1)^T.$$

Since B is normal,

$$H = B^* B - B B^* + cc^* = cc^* \geq 0, \qquad \operatorname{rank} H = 1.$$

However, the active spectrum is $\{0, 1, i\}$, which is not collinear, so Corollary 3 rules out a normal completion.

For a direct algebraic check, normality would force $d = \omega c$ with $\omega \in \mathbb{T}$ and

$$\lambda - \omega \overline{\lambda} = \eta - \omega \overline{\eta} \quad (\lambda \in \{0, 1, i\}).$$

The equation for $\lambda = 0$ makes the common value zero. The equation for $\lambda = 1$ then forces $\omega = 1$, while the equation for $\lambda = i$ forces $\omega = -1$, a contradiction. $\square$

The source already observes that its positive-semidefinite defect condition is not sufficient for general square border blocks of size at least three. Proposition 4 sharpens the phenomenon to a single missing column and gives the complete geometric reason in the normal-B case. Within this family, three active noncollinear eigenvalues are necessary, so dimension four is minimal.

6 Verification notes

- The proof uses only compact SVD, direct block multiplication, and the spectral theorem for the normal-compression corollary.
- The potentially delicate $n = m$ case is covered by $\ker X \subseteq \ker Y$: it gives $Y = YW_1W_1^*$ even though W_1 is not square.
- The trace of the upper-left defect identity gives $\|d\| = \|c\|$, so no independent bottom-right condition is missing.
- The deterministic script `code/verify_codim_one.py` checks the two block residuals on a random unitarily diagonalized normal matrix, constructs a non-Hermitian positive example with collinear active spectrum, and verifies the rank-one defect and incompatible phases in Proposition 4.

7 Limitations and novelty audit

This is a partial result: it does not characterize normal targeting when $\operatorname{codim} \operatorname{Ran} X \geq 2$. In the nonnormal-$B$ case, the theorem leaves a one-phase compatibility test rather than eliminating that phase by a closed formula.

A bounded search on 27 August 2026 used the exact arXiv id and title, the phrases "normal targeting", "normal completion one missing column", and "normal defect one", and the authors' names. It found the source paper and the established normal-defect-one literature, especially Kaliuzhnyi-Verbovetskyi, Spitkovsky, and Woerdeman [2]. That work classifies completions of a prescribed north-west block when both border vectors may vary. Our targeting reduction instead prescribes the lower border c^* and leaves only the upper border and corner free. No source found in the bounded search stated the rank-$m-1$ targeting criterion, the active spectrum affine-line criterion, or the explicit 4×4 obstruction above. Absolute novelty is not claimed; the result is close in spirit to normal completion theory and may be derivable from a broader formulation not found in the search.

8 Human-review recommendation

Recommend expert review as a likely valid partial solution. The two points most worth checking are (1) the row-space implication $Y = YW_1W_1^*$ under $\ker X \subseteq \ker Y$, and (2) the conjugations in equation (1). If those pass, the theorem gives an exact and constructive solution of the maximal-rank singular subcase and the four-dimensional example is a rigorous obstruction, not a numerical claim.

References

[1] K. Bierly, S. R. Garcia, and R. A. Horn, *The linear targeting problem*, arXiv:2408.10036, 2024 (revised 2025).

[2] D. S. Kaliuzhnyi-Verbovetskyi, I. M. Spitkovsky, and H. J. Woerdeman, *Matrices with normal defect one*, Operators and Matrices **3** (2009), 427–450; arXiv:0903.0090.